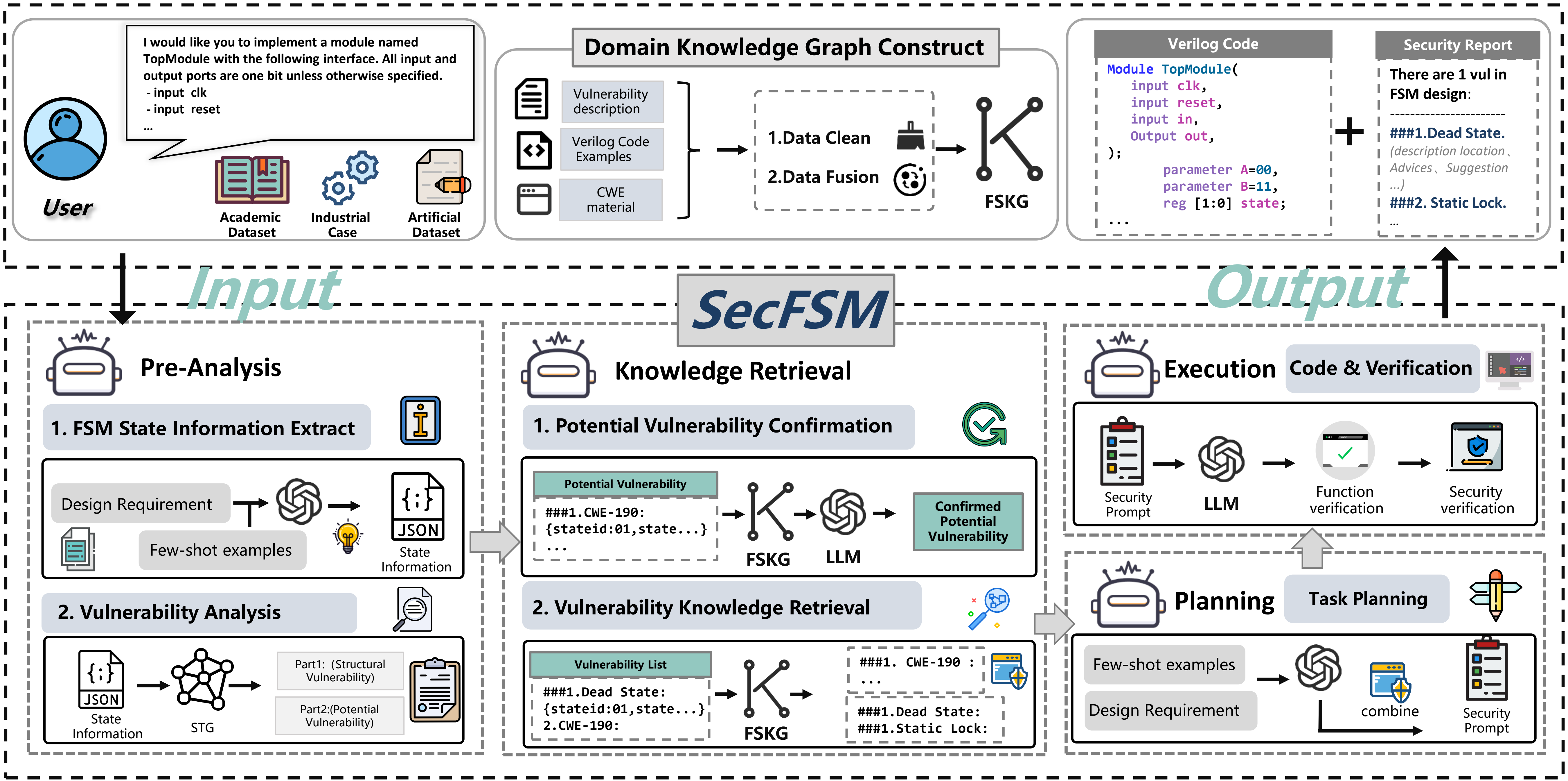
I would like you to implement a module named TopModule with the following interface. All input and output ports are one bit unless otherwise specified.
- input clk
- input reset
...
User
Academic Dataset
Industrial Case
Artificial Dataset
Domain Knowledge Graph Construct
Vulnerability description
Verilog Code Examples
CWE material
1.Data Clean
2.Data Fusion
FSKG
Verilog Code
Module TopModule(
input clk,
input reset,
input in,
Output out,
);
parameter A=00,
parameter B=11,
reg [1:0] state;
...
Security Report
There are 1 vul in FSM design:
###1.Dead State.
(description location、Advices、Suggestion ...)
###2. Static Lock.
...
Input
SecFSM
Output
Pre-Analysis
1. FSM State Information Extract
Design Requirement
Few-shot examples
JSON
State Information
2. Vulnerability Analysis
JSON
State Information
STG
Part1: (Structural Vulnerability)
Part2:(Potential Vulnerability)
Knowledge Retrieval
1. Potential Vulnerability Confirmation
Potential Vulnerability
###1.CWE-190:
{stateid:01,state...}
...
FSKG
LLM
Confirmed Potential Vulnerability
2. Vulnerability Knowledge Retrieval
Vulnerability List
###1.Dead State:
{stateid:01,state...}
2.CWE-190:
FSKG
###1. CWE-190 :
...
###1.Dead State:
###1.Static Lock:
Execution
Code & Verification
Security Prompt
LLM
Function verification
Security verification
Planning
Task Planning
Few-shot examples
Design Requirement
combine
Security Prompt